

E-Commerce Website Project Documentation

Technology Stack: React / Next.js, Tailwind CSS, Material UI or ShadCN, Firebase or Supabase

1. Project Overview

This project is a fully functional eCommerce website that allows users to browse products, add items to the cart, and place orders using Cash on Delivery (COD). The system includes two main roles: Client (customer) and Admin. The platform will provide a modern UI, secure authentication, and an admin dashboard for managing products and orders.

2. Technology Stack

- Frontend: React.js or Next.js
- Styling: Tailwind CSS
- Component Library: Material UI (React) or ShadCN (Next.js)
- Backend / Database: Firebase or Supabase
- Authentication: Firebase Auth / Supabase Auth
- Hosting: Vercel / Netlify

3. User Roles

3.1 Client (Customer)

- View all products without login
- Add products to cart without login
- User must login or signup to place an order
- Fill checkout form including name, phone number, address, city, etc.
- Place order using Cash on Delivery
- View their previous orders (optional feature)

3.2 Admin

- Admin login authentication
- Admin dashboard overview
- Add new products
- Edit existing products
- Delete products
- Create and manage product categories
- View all orders
- Update order status (Pending, Processing, Delivered)

- View monthly order statistics
- View total revenue analytics

4. Website Pages (Client Side)

- Home Page – Featured products, categories, promotions
- Product Listing Page – Display all products
- Product Detail Page – Product description, price, images
- Cart Page – Display selected products
- Checkout Page – Address form and order confirmation
- Login / Signup Page – User authentication

5. Admin Dashboard Pages

- Dashboard Overview (Orders, Revenue, Statistics)
- Product Management Page
- Add / Edit Product Page
- Category Management Page
- Orders Management Page
- Revenue Analytics Page

6. Database Structure (Suggested)

Collection / Table	Fields
Users	id, name, email, role (admin/client), createdAt
Products	id, title, description, price, categoryId, images, stock
Categories	id, name, createdAt
Orders	id, userId, products, totalPrice, address, phone, status, createdAt

7. Order Workflow

- User browses products
- User adds products to cart
- User logs in or signs up
- User fills checkout form
- Order is saved in database with 'Pending' status
- Admin views order in dashboard
- Admin updates order status (Processing / Delivered)

8. UI Requirements

- Modern responsive design
- Mobile friendly layout
- Clean product cards
- Dashboard charts for analytics
- Proper spacing and professional layout

9. Features

- Product search
- Product filters (price, category)
- User order history
- Admin export orders